Learning Reliability-Aware Audio-Visual Fusion for Multimodal Retrieval

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Abstract

Multimodal retrieval over video content requires jointly reasoning over visual scenes and spoken language to identify semantically relevant matches from large candidate sets. Existing practical systems often process each modality independently and reduce both to text before retrieval, which weakens cross-modal alignment and discards fine-grained visual information. In this paper, we propose a reliability-aware audio-visual fusion framework for multimodal retrieval that learns a shared representation from visual keyframes and aligned speech transcripts. The method combines direct visual embeddings from pretrained vision-language models with transcript-based language embeddings and uses a gated fusion mechanism to adaptively weight each modality according to its informativeness and quality. This design allows the model to remain robust when one modality is noisy, incomplete, or less discriminative. We formulate retrieval as a ranking problem in a joint embedding space and evaluate the framework using standard retrieval metrics, including Recall@K, mean Average Precision, and nDCG. We compare against strong unimodal and multimodal baselines, including CLIP-based visual retrieval, transcript-only retrieval, and late-fusion approaches. In addition, we study a multilingual setting to analyze how improved Indian-language speech recognition affects downstream retrieval quality in dialogue-heavy video content. The proposed framework is designed to provide a stronger foundation for multimodal similarity search by preserving modality-specific information while learning when and how to fuse complementary evidence.

Keywords: Multimodal Retrieval, Audio-Visual Fusion, Similarity Search, Video Retrieval, Cross-Modal Representation Learning, Contrastive Learning

1 Introduction

Multimodal retrieval has become an increasingly important problem in modern AI systems as digital content continues to grow in scale, diversity, and complexity. Large video collections contain rich information distributed across multiple modalities, including visual scenes, spoken language, background audio, and temporal context. Retrieving semantically relevant content from such data cannot be addressed effectively through a single modality alone. Two videos may appear visually similar while conveying entirely different meaning through dialogue, and conversely, similar spoken content may occur in visually distinct scenes. This makes multimodal retrieval a fundamental problem for applications such as video search, contextual content matching, recommendation, and advertisement placement.

Recent advances in vision-language modeling and contrastive representation learning have significantly improved retrieval performance by mapping images and text into a shared semantic space. Models such as CLIP and related vision-language architectures have shown that large-scale pretraining can produce highly transferable visual representations for retrieval tasks. At the same time, progress in automatic speech recognition has made it possible to extract semantic information from spoken content at scale. Despite these advances, many practical systems for multimedia similarity search still follow a pipeline-based design in which visual frames are first converted into captions, audio is transcribed into text, and both modalities are then reduced to textual embeddings for downstream matching. Although this approach is simple and convenient, it introduces important limitations. Converting visual information entirely into text can discard fine-grained scene details, heuristic fusion fails to account for varying modality quality, and the retrieval objective is often not optimized directly for multimodal ranking.

These limitations are particularly important in real-world video retrieval settings, where the usefulness of a modality depends heavily on context. In some clips, visual appearance may be highly informative while the spoken content is sparse or noisy. In others, dialogue may contain the most discriminative semantics even when the visuals are generic. A robust multimodal retrieval system should therefore not only preserve modality-specific information, but also learn when to trust one modality more than the other. This motivates a retrieval-oriented fusion framework that explicitly models the complementary and variable reliability of visual and spoken signals.

In this paper, we reformulate multimodal similarity search as a learning problem over a shared audio-visual embedding space. Instead of reducing both modalities to text alone, we preserve direct visual semantics by encoding keyframes using pretrained vision-language backbones and represent spoken content through transcript-based language embeddings. We then introduce a reliability-aware gated fusion mechanism that adaptively combines the two modalities according to their informativeness and estimated quality. The resulting fused representation is optimized for retrieval and used to rank candidate items through similarity search in a joint embedding space. This formulation enables the model to remain robust when one modality is noisy, incomplete, or less discriminative, while still benefiting from complementary cross-modal evidence.

We study this problem in the context of multimodal video retrieval, with MSR-VTT serving as the primary benchmark and a secondary multilingual setting used to

analyze the effect of speech recognition quality on downstream retrieval performance. The multilingual setting is particularly relevant for dialogue-heavy video content, where transcript quality can substantially affect retrieval quality. By including this secondary analysis, we examine not only whether multimodal fusion improves ranking performance, but also how improved speech recognition for Indian languages influences retrieval in practice.

The main contributions of this work are threefold. First, we propose a reliability-aware audio-visual fusion framework for multimodal retrieval that combines keyframe-based visual embeddings and transcript-based language embeddings in a shared retrieval space. Second, we formulate multimodal retrieval as a ranking problem and evaluate it using standard retrieval metrics, including Recall@K, mean Average Precision, and nDCG, against strong unimodal and multimodal baselines. Third, we provide a multilingual analysis showing how transcript quality affects retrieval in dialogue-driven settings, offering additional insight into the interaction between speech recognition and multimodal similarity search.

Overall, this work aims to move beyond pipeline-style multimedia matching systems toward a more principled multimodal retrieval formulation. By preserving direct modality-specific information and learning how to fuse complementary evidence, the proposed approach provides a stronger foundation for scalable and semantically meaningful retrieval over video content.

2 Related Work

Multimodal retrieval seeks to learn representations that support semantic matching across heterogeneous inputs such as text, images, audio, and video. Recent progress in this area has been driven by shared embedding approaches, especially contrastive learning frameworks that align paired samples in a common representation space. In the video domain, this line of work has led to strong retrieval systems that jointly model video and language for text-to-video and video-to-text search. For example, Frozen in Time formulates video-text retrieval through a joint encoder trained on image and video caption data, while VideoCLIP extends contrastive pretraining to temporally aligned video-text pairs for zero-shot video understanding and retrieval [1, 2]. These works demonstrate that retrieval performance improves substantially when multimodal structure is learned directly rather than engineered through independent pipelines.

Standardized benchmarks have also played an important role in shaping this literature. In particular, MSR-VTT remains one of the most widely used benchmarks for video-language retrieval and video understanding, providing a large and diverse collection of video clips paired with natural language descriptions [3]. Its widespread adoption makes it a suitable benchmark for evaluating retrieval-oriented multimodal systems and comparing against prior work.

Vision-language pretraining has become the dominant paradigm for semantic retrieval in image-text and video-text settings. A particularly influential milestone was CLIP, which showed that contrastive pretraining on a very large corpus of image-text pairs yields transferable visual representations and a shared embedding space suitable

for zero-shot transfer and retrieval [4]. CLIP is especially relevant to retrieval because it directly learns embeddings optimized for cross-modal similarity rather than relying on intermediate caption generation.

Subsequent models have refined this paradigm in different ways. SigLIP replaces the standard softmax-normalized contrastive loss with a pairwise sigmoid objective, which improves training efficiency and supports strong language-image pretraining performance [5]. BLIP and BLIP-2 further explore vision-language pretraining from the perspective of representation learning and generation, with BLIP-2 in particular showing that strong vision-language performance can be achieved by combining frozen image encoders with large language models through lightweight trainable adapters [6, 7]. These models provide a natural set of baselines and building blocks for multimodal retrieval systems that require robust visual encoders.

At the same time, caption-centric pipelines remain common in practical multimedia systems. However, reducing visual information to generated captions can discard fine-grained scene attributes, object relationships, and contextual cues that are preserved in direct visual embeddings. For retrieval tasks, this makes pretrained visual encoders such as CLIP or SigLIP particularly attractive compared with purely caption-based matching pipelines [4, 5]. This motivates our decision to treat direct visual embeddings as the primary visual signal and use caption-based methods only as comparison baselines.

Spoken language is often a major source of semantic information in videos, especially in dialogue-heavy, instructional, or commentary-driven content. Modern automatic speech recognition systems enable spoken content to be converted into text for downstream indexing and retrieval, making transcript-aware search increasingly practical. Whisper is particularly relevant in this context because it was trained on large-scale multilingual and multitask supervision and was shown to generalize strongly across languages and noise conditions in a zero-shot setting [8]. This makes it a strong starting point for transcript-based retrieval and multilingual audio analysis.

Despite this progress, the downstream impact of transcript quality on multimodal retrieval remains underexplored. Errors in speech recognition can distort semantic meaning and weaken alignment between spoken and visual content, particularly in multilingual settings or underrepresented languages. This motivates our secondary analysis of how improved speech recognition quality affects retrieval performance in Indian-language and dialogue-heavy video settings. Rather than treating ASR as an isolated preprocessing step, our formulation views transcript quality as part of the multimodal retrieval problem itself [8].

A central challenge in multimodal learning is how to combine modality-specific signals without suppressing complementary information or over-relying on noisy inputs. Prior work has explored early fusion, late fusion, attention-based interaction, and gating-based mechanisms. Among these, gated fusion is especially relevant when the informativeness of each modality varies across examples. Gated Multimodal Units introduced a neural gating mechanism that learns how different modalities should influence an intermediate representation, showing improvements over single-modality and alternative fusion baselines [9]. This idea is directly aligned with retrieval settings where visual and spoken signals can differ substantially in reliability from one sample to another.

In retrieval, simple late fusion remains common because it is easy to implement and computationally efficient. However, fixed-weight or heuristic fusion strategies cannot adapt to per-sample modality quality. In video retrieval, this is a serious limitation because some clips are visually distinctive but linguistically weak, while others are semantically driven by dialogue despite visually generic scenes. This motivates reliability-aware fusion, where the contribution of each modality is learned dynamically rather than chosen heuristically. Our method builds on this intuition by using a gated audio-visual fusion mechanism specifically designed for retrieval in a shared embedding space [9].

The present work lies at the intersection of video-language retrieval, speech-aware video understanding, and multimodal fusion. It is most closely related to retrieval frameworks that use pretrained vision-language models for semantic alignment and to multimodal learning methods that adaptively combine heterogeneous signals. However, unlike practical pipeline-style systems that reduce both modalities entirely to text before matching, our formulation preserves direct visual semantics through pretrained visual embeddings and models spoken content through transcript-based language embeddings. In addition, instead of relying on fixed or heuristic fusion, we introduce a reliability-aware gating mechanism that adaptively weights visual and spoken signals during retrieval.

By framing the task explicitly as multimodal retrieval, the present work aligns more naturally with benchmark-driven evaluation using datasets such as MSR-VTT and metrics such as Recall@K, mean Average Precision, and nDCG [3]. This positioning enables stronger comparison against modern baselines including CLIP, SigLIP, BLIP-2, transcript-only retrieval, and naive late-fusion systems, while also allowing us to study how transcript quality affects retrieval under multilingual conditions.

3 Methodology

3.1 Overview

We propose a *reliability-aware audio-visual fusion framework* for multimodal retrieval over video content. The objective is to learn a shared representation that captures complementary semantic information from visual scenes and spoken language while remaining robust when one modality is noisy, incomplete, or weakly informative. Given a query video and a database of candidate items, the system ranks candidates according to similarity in a learned joint embedding space.

The framework consists of four main stages:

1. visual keyframe extraction and encoding,
2. speech transcription and language encoding,
3. reliability-aware multimodal fusion, and
4. similarity-based retrieval in a shared embedding space.

Unlike earlier pipeline-based systems that reduce both modalities entirely to text, the proposed approach preserves direct visual semantics through pretrained visual encoders and models spoken content separately through transcript-based language

representations. The two modalities are then fused through a gated mechanism that adaptively determines their relative contribution for each sample.

3.2 Problem Formulation

Let a multimodal video item be represented as

$$x = \left(x^{(v)}, x^{(a)}\right), \tag{1}$$

where $x^{(v)}$ denotes the visual stream and $x^{(a)}$ denotes the corresponding audio stream. For retrieval, each item is mapped into a shared multimodal embedding space

$$\mathcal{S} \subseteq \mathbb{R}^d \tag{2}$$

through an embedding function $g(\cdot)$, such that

$$g(x) = \mathbf{z} \in \mathcal{S}. \tag{3}$$

Given a query item q and a candidate database $\mathcal{D} = \{d_1, d_2, \ldots, d_N\}$, the retrieval problem is to rank all candidates according to their semantic similarity with the query. We define the similarity search function as

$$f(x, y) = \mathrm{sim}(g(x), g(y)), \tag{4}$$

where $\mathrm{sim}(\cdot, \cdot)$ is cosine similarity:

$$f(x, y) = \frac{g(x)^\top g(y)}{\|g(x)\|_2 \, \|g(y)\|_2}. \tag{5}$$

A higher value of $f(x, y)$ indicates stronger semantic relevance between the two multimodal items.

3.3 Visual Representation Learning

The visual modality is modeled using keyframes extracted from the video stream. Since full video frame processing is computationally expensive and often redundant, each video is represented through a subset of informative frames obtained either through uniform temporal sampling or shot-based keyframe extraction. Let the set of extracted frames for a video be

$$F = \{f_1, f_2, \ldots, f_T\}. \tag{6}$$

Each keyframe is encoded using a pretrained vision-language model such as CLIP or SigLIP [4, 5]. The encoder maps each frame f_t to a visual embedding:

$$\mathbf{v}_t = E_v(f_t). \tag{7}$$

To obtain a single visual representation for the video, frame-level embeddings are aggregated across time. In the simplest form, this is done using average pooling:

$$\mathbf{v} = \frac{1}{T} \sum_{t=1}^{T} \mathbf{v}_t. \tag{8}$$

This aggregated embedding preserves direct visual semantics without forcing the system to rely on caption generation. Compared with caption-based representations, direct visual embeddings retain more fine-grained information about scene composition, objects, and visual context.

3.4 Audio and Transcript Representation

The audio modality is modeled through speech transcription followed by language encoding. Given the audio stream $x^{(a)}$, a transcript is first obtained using an automatic speech recognition system such as Whisper [8]. The resulting transcript is denoted by

$$\tau = \mathrm{ASR}\left(x^{(a)}\right). \tag{9}$$

The transcript is then encoded using a text encoder to produce a language embedding:

$$\mathbf{a} = E_a(\tau). \tag{10}$$

This representation captures semantic information conveyed through spoken dialogue, narration, or commentary. It is especially useful in cases where the visual scene is generic but the speech content is highly informative. In multilingual settings, transcript quality depends strongly on ASR performance, making this branch particularly sensitive to language-specific noise and recognition errors.

3.5 Projection into a Shared Retrieval Space

The visual and transcript representations are initially produced in modality-specific feature spaces. To align them within a common retrieval space, learned linear projections are applied:

$$\tilde{\mathbf{v}} = P_v \mathbf{v}, \qquad \tilde{\mathbf{a}} = P_a \mathbf{a}, \tag{11}$$

where $\tilde{\mathbf{v}}, \tilde{\mathbf{a}} \in \mathbb{R}^d$.

This step ensures that both modalities are represented in a comparable latent space before fusion. The projections also allow the model to adapt pretrained representations to the retrieval objective rather than using them in a completely frozen and task-agnostic way.

3.6 Reliability-Aware Gated Fusion

A key challenge in multimodal retrieval is that the relative informativeness of each modality varies across samples. Some videos are visually distinctive but contain little

useful spoken content, while others are semantically defined by dialogue despite visually generic frames. To model this variation, we introduce a reliability-aware gating mechanism inspired by prior gated multimodal fusion methods [9].

For the visual and transcript representations, scalar reliability weights are estimated as

$$\alpha_v = \sigma\big(\mathbf{w}_v^\top \tilde{\mathbf{v}} + b_v\big)\,, \qquad \alpha_a = \sigma\big(\mathbf{w}_a^\top \tilde{\mathbf{a}} + b_a\big)\,, \tag{12}$$

where $\sigma(\cdot)$ denotes the sigmoid activation function and $\alpha_v, \alpha_a \in [0, 1]$.

These weights control how much each modality contributes to the final fused representation. The fused multimodal embedding is computed as

$$\mathbf{z} = \mathrm{Norm}(\alpha_v \tilde{\mathbf{v}} + \alpha_a \tilde{\mathbf{a}})\,, \tag{13}$$

where $\mathrm{Norm}(\cdot)$ denotes normalization, such as ℓ_2-normalization.

This formulation allows the model to emphasize visual evidence when the transcript is noisy or weak, and conversely to rely more on spoken semantics when the visual signal is less discriminative. Compared with fixed-weight late fusion, the gated design provides input-adaptive fusion and improves robustness across diverse video conditions.

3.7 Retrieval Objective

The fused representation is trained for retrieval using a contrastive ranking objective. Let (x_i, y_i^+) denote a positive query-target pair and let y_j^- denote negative examples. The objective is to make the similarity of positive pairs larger than that of negative pairs in the shared embedding space.

A standard contrastive formulation is the InfoNCE loss:

$$\mathcal{L}_{\mathrm{NCE}} = -\log \frac{\exp\big(f(x_i, y_i^+)/\tau\big)}{\exp\big(f(x_i, y_i^+)/\tau\big) + \sum_j \exp\big(f(x_i, y_j^-)/\tau\big)}, \tag{14}$$

where τ is a temperature parameter.

This objective encourages semantically related items to cluster together while separating irrelevant pairs. Because the retrieval function is learned directly from similarity comparisons, the resulting embedding space is more suitable for ranking than a pipeline based solely on post hoc cosine similarity over independently engineered features.

3.8 Similarity Search and Ranking

After training, all items in the database are embedded into the shared multimodal space $\mathcal{S}$. Given a query embedding $g(q)$, candidates are ranked using the similarity function $f(q, d_i)$. The top-ranked candidates are returned as the retrieval results:

$$\mathrm{rank}(q) = \mathrm{sort}_{d_i \in \mathcal{D}}\ f(q, d_i). \tag{15}$$

For efficiency at scale, the learned embeddings can be indexed using approximate nearest-neighbor search techniques. This makes the framework suitable not only for

benchmark evaluation but also for practical retrieval systems operating over large multimodal databases.

3.9 Design Rationale

The proposed methodology is motivated by three considerations. First, direct visual embeddings preserve scene-level semantics more effectively than caption-only pipelines, making them a stronger foundation for retrieval. Second, transcript-based representations capture spoken content that is often invisible in visual frames, particularly in dialogue-heavy videos. Third, reliability-aware fusion provides a principled way to combine these modalities under varying quality conditions, avoiding the limitations of fixed or heuristic fusion strategies.

Taken together, these components produce a retrieval-oriented framework that is both conceptually grounded and practically scalable. By learning when and how to combine visual and spoken evidence, the method aims to improve multimodal similarity search in settings where either modality alone is insufficient.

4 Experiments

4.1 Experimental Setup

4.1.1 Datasets

We evaluate the proposed framework on a standard multimodal retrieval benchmark and a secondary multilingual setting. As the primary benchmark, we use **MSR-VTT**, which is widely adopted for video-language retrieval and provides a large collection of short video clips paired with natural language descriptions [3]. This benchmark enables direct comparison with prior retrieval systems and provides a standardized setting for evaluating ranking quality.

To further analyze the effect of speech quality on multimodal retrieval, we also consider a secondary multilingual audio-video setting focused on dialogue-heavy content. This setting is used to study how transcript quality, particularly for Indian-language speech, affects downstream retrieval performance. The multilingual analysis is not treated as the primary benchmark, but rather as an additional evaluation setting that highlights the interaction between automatic speech recognition and multimodal retrieval.

For the main benchmark, videos are split into training, validation, and test sets following the standard protocol of the dataset. In the multilingual setting, the data is partitioned into disjoint train, validation, and test subsets to ensure that evaluation is performed on unseen samples.

4.1.2 Input Processing

Each video is represented using both visual and spoken information.

For the **visual modality**, a set of keyframes is extracted from each video using either uniform temporal sampling or shot-based selection. The extracted frames are encoded using a pretrained vision-language model such as CLIP, with SigLIP included

as an additional strong baseline [4, 5]. Frame-level embeddings are then aggregated to obtain a single visual representation per video.

For the **audio modality**, the speech signal is transcribed using Whisper [8]. The resulting transcript is encoded using a text encoder to produce the language representation. In the multilingual analysis, a stronger or fine-tuned speech recognition model may be used to evaluate the effect of transcript quality on retrieval.

4.1.3 Model Variants

We compare the proposed reliability-aware fusion model against several baselines representing different levels of multimodal integration.

The evaluated systems include:

1. **Visual-only retrieval**, where only the visual encoder is used.
2. **Transcript-only retrieval**, where only the transcript embedding is used.
3. **Caption-based retrieval**, where visual frames are converted to text captions and retrieval is performed in the text space.
4. **Late-fusion retrieval**, where visual and transcript similarity scores are combined using fixed weights.
5. **Concatenation-based fusion**, where modality embeddings are concatenated before projection.
6. **Proposed reliability-aware gated fusion**, where visual and transcript representations are adaptively weighted before fusion.

In addition to CLIP-based visual encoding, we include **SigLIP** as a stronger visual baseline, and optionally **BLIP-2** as an auxiliary comparison model when caption-based or richer vision-language representations are considered [6, 7].

4.1.4 Training Protocol

The retrieval model is trained using a contrastive ranking objective over positive and negative pairs. Positive pairs correspond to semantically matched query-target items, while negatives are sampled from the remaining items in the batch or from the candidate pool. The training objective is the InfoNCE loss defined in Section **??**.

The visual projection layer, audio projection layer, and gating parameters are optimized jointly. Depending on computational constraints, the pretrained backbone encoders may either remain frozen or be partially fine-tuned. In the default setting, the pretrained encoders are kept fixed and only the projection and fusion layers are trained, which reduces computational cost and stabilizes optimization.

Training is performed using the AdamW optimizer with a fixed learning rate schedule. Hyperparameters including batch size, learning rate, projection dimension, and temperature parameter are selected using validation-set performance. Early stopping is applied based on retrieval quality on the validation split.

4.1.5 Evaluation Metrics

We evaluate retrieval performance using standard ranking metrics:

- **Recall@K (R@1, R@5, R@10)**, which measures how often a relevant item appears within the top K retrieved results,
- **mean Average Precision (mAP)**, which captures ranking quality across the full retrieved list,
- **normalized Discounted Cumulative Gain (nDCG)**, which emphasizes the quality of higher-ranked items.

These metrics are standard in retrieval evaluation and provide a more meaningful assessment than raw similarity scores alone. In the multilingual analysis, we additionally report **Word Error Rate (WER)** to quantify transcript quality and relate it to downstream retrieval performance.

4.2 Baselines

A rigorous evaluation of multimodal retrieval requires comparison against both unimodal and multimodal baselines. Accordingly, we compare the proposed method against the following systems:

- **CLIP visual-only retrieval** [4],
- **SigLIP visual-only retrieval** [5],
- **Transcript-only retrieval**,
- **Caption-based retrieval**,
- **Late-fusion retrieval**,
- **Concatenation-based fusion**,
- **Proposed reliability-aware gated fusion**.

These baselines are chosen to isolate the effect of direct visual embeddings, transcript information, and adaptive fusion. In particular, comparing against caption-based retrieval highlights the benefit of preserving direct visual semantics, while comparison with late fusion demonstrates whether adaptive gating provides a measurable advantage over fixed fusion strategies.

4.3 Main Results

The main results compare all methods on the primary benchmark using Recall@K, mAP, and nDCG. The central question is whether reliability-aware fusion improves retrieval quality over unimodal and naive multimodal baselines.

We expect three broad trends. First, direct visual embeddings from strong pretrained vision-language models should outperform caption-based retrieval, since they preserve more fine-grained scene information. Second, transcript-only retrieval should perform well in dialogue-driven settings but may underperform when visual semantics dominate. Third, the proposed fusion model should outperform both unimodal baselines and fixed-weight late fusion by adaptively combining the most informative modality for each input.

A typical state-of-the-art comparison table should report the visual encoder, transcript encoder, fusion strategy, and retrieval metrics for each method. The proposed model should be evaluated under the same data split and metric definitions as the baselines to ensure fairness.

4.4 Ablation Studies

Ablation experiments are critical for understanding which components of the proposed framework contribute most to retrieval performance. We consider the following ablations:

1. **Visual-only vs. audio-only vs. fused retrieval.** This comparison measures the standalone utility of each modality and quantifies the gain from multimodal fusion.
2. **Late fusion vs. gated fusion.** This ablation isolates the impact of the reliability-aware gating mechanism.
3. **Direct visual embeddings vs. caption-based retrieval.** This comparison tests whether retaining direct visual representations provides a retrieval advantage over converting frames to text.
4. **CLIP vs. SigLIP visual encoders.** This evaluates whether the proposed fusion method is robust across different pretrained visual backbones.
5. **Effect of transcript quality.** In the multilingual setting, we compare retrieval performance using baseline Whisper transcripts and improved transcripts to measure how ASR quality affects downstream retrieval.
6. **Missing or noisy modality analysis.** We simulate degraded transcripts or incomplete frame inputs to test robustness when one modality is unreliable.

These ablations provide a more complete picture of how the model behaves and help determine whether the gains come from the fusion mechanism itself or simply from stronger encoders.

4.5 Qualitative Analysis

In addition to quantitative metrics, qualitative retrieval examples are used to illustrate the behavior of the model under different modality conditions. We examine representative cases where:

- the visual modality alone retrieves incorrect but visually similar results,
- the transcript modality alone retrieves semantically related but visually mismatched results,
- the fused model correctly retrieves items by combining both signals.

Such examples are especially useful for showing how the proposed gating mechanism resolves ambiguity. For instance, a visually generic indoor scene may only become distinguishable when dialogue is taken into account, whereas a silent but visually distinctive scene may rely almost entirely on visual content. Qualitative analysis therefore complements the aggregate metrics by demonstrating how multimodal reasoning improves retrieval in practice.

4.6 Discussion

The experimental design is intended to test whether multimodal retrieval benefits from learning an input-adaptive fusion of visual and spoken semantics rather than relying on fixed or heuristic pipelines. Strong performance on the primary benchmark would demonstrate that the proposed approach is effective under standard retrieval

evaluation, while the multilingual analysis would provide additional evidence that transcript quality plays an important role in retrieval quality when spoken content carries substantial semantic load.

Overall, the experiments are designed not only to compare against state-of-the-art baselines, but also to justify the methodological choices made in the proposed framework. The inclusion of unimodal comparisons, fusion ablations, and multilingual analysis ensures that the evaluation directly addresses the core claims of the paper.

5 Conclusion

In this work, we reformulated multimodal similarity search as a retrieval problem over a shared audio-visual embedding space. Rather than reducing both modalities entirely to text, the proposed framework preserves direct visual semantics through pretrained vision-language encoders and models spoken content through transcript-based language representations. To combine these complementary signals, we introduced a reliability-aware gated fusion mechanism that adaptively weights the contribution of each modality based on its informativeness and quality.

The proposed method is designed to address a core limitation of pipeline-style multimedia retrieval systems, namely their reliance on fixed or heuristic fusion strategies that do not account for per-sample variation in modality quality. By learning a fused representation optimized directly for ranking, the framework provides a more principled approach to multimodal retrieval over video content. The experimental design further supports this objective through comparison against strong unimodal and multimodal baselines, evaluation on a standard benchmark, and additional multilingual analysis of transcript quality.

Overall, the work aims to provide a stronger foundation for multimodal retrieval by unifying visual and spoken semantics within a retrieval-oriented representation learning framework. We believe this direction is particularly promising for large-scale video search, contextual content matching, and downstream applications where either visual or spoken information alone is insufficient to capture semantic relevance.

6 Limitations

Despite its advantages, the proposed framework has several limitations. First, the method depends on the quality of the underlying pretrained encoders. Retrieval performance may degrade when the visual backbone fails to capture domain-specific semantics or when automatic speech recognition introduces significant transcription errors. This issue is especially relevant in multilingual and low-resource language settings, where transcript quality can vary considerably.

Second, the current formulation represents the audio modality primarily through speech transcripts. While this is effective for dialogue-heavy content, it does not explicitly model non-speech acoustic signals such as background sounds, music, or environmental audio cues, which may also carry useful retrieval information. Incorporating raw acoustic representations or dedicated audio encoders could improve performance in settings where non-verbal sound is semantically important.

Third, the fusion mechanism is intentionally lightweight in order to preserve scalability, but this design may limit the model's ability to capture deeper cross-modal interactions. More expressive architectures, such as cross-attention or transformer-based fusion, may yield stronger performance at the cost of additional computational complexity.

Finally, the multilingual evaluation is intended as a secondary analysis rather than a primary benchmark. As a result, conclusions regarding the impact of transcript quality on retrieval should be interpreted with appropriate caution until validated on larger and more standardized multilingual datasets. Future work should therefore investigate broader multilingual benchmarks, stronger acoustic modeling, and richer cross-modal fusion strategies.

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